

ALTUS AFB, OK, USA

WMO: 723520

Lat: 34.650N

Lon: 99.267W

Elev: 421

StdP: 96.37

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13902

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.5	-7.0	-17.0	0.9	-3.4	-14.4	1.1	-2.1	14.1	5.2	12.6	6.0	4.1	20	0.568

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.0	40.4	22.7	38.8	22.8	37.3	22.8	25.5	34.3	24.9	33.5	24.2	32.6	5.6	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.4	19.2	27.8	22.7	18.4	27.4	22.2	17.8	27.2	80.8	34.2	77.7	33.3	75.2	32.5	28.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.4	11.1	9.8	-13.6	42.0	2.7	2.1	-15.5	43.5	-17.0	44.8	-18.6	46.0	-20.5	47.5		
(5)			-14.3	26.3	2.4	1.1	-16.0	27.1	-17.4	27.7	-18.8	28.3	-20.6	29.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.1	4.6	7.2	11.9	16.2	21.9	27.1	29.0	28.5	24.2	17.6	10.9	5.3		
	DBStd	9.82	5.30	5.94	5.69	4.62	4.37	3.22	2.72	2.65	3.88	4.87	4.92	5.00		
	HDD10.0	556	179	112	46	7	0	0	0	0	0	6	47	161		
	HDD18.3	1767	426	313	208	92	17	0	0	0	6	73	227	405		
	CDD10.0	3145	12	33	105	193	369	514	589	575	427	239	73	14		
	CDD18.3	1313	0	1	9	28	128	264	331	317	183	49	3	0		
	CDH23.3	16517	11	43	162	448	1471	3316	4501	3987	1931	574	67	7		
	CDH26.7	9097	1	10	45	163	690	1875	2742	2364	987	206	12	1		
(14) Wind	WSAvg	4.7	4.5	4.9	5.4	5.7	5.2	5.1	4.1	3.9	4.1	4.4	4.4	4.2		
(15) Precipitation	PrecAvg	689	24	30	43	53	104	101	50	71	79	65	39	28		
	PrecMax	1078	105	115	124	188	339	280	190	188	255	299	184	123		
	PrecMin	265	0	0	0	0	1	3	0	0	1	0	0	0		
	PrecStd	185	26	27	32	40	73	65	44	57	61	66	40	29		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.0	27.8	31.0	34.1	38.2	42.2	42.3	42.2	38.9	33.9	28.4	23.8		
		MCWB	12.9	14.6	16.5	17.0	20.6	22.4	22.4	22.3	21.7	19.8	17.2	13.4		
	2%	DB	20.7	24.0	27.4	30.9	35.4	39.1	40.4	39.9	36.5	31.3	25.3	19.5		
		MCWB	10.7	13.3	16.2	17.3	20.8	22.5	23.1	22.9	22.0	19.6	15.7	11.0		
	5%	DB	17.7	20.9	24.8	28.1	33.0	37.0	38.8	38.0	34.6	28.9	22.6	16.5		
		MCWB	9.7	11.7	15.4	16.9	20.6	22.6	23.1	23.2	21.7	18.8	14.2	10.5		
	10%	DB	14.6	17.6	22.1	25.7	30.6	35.0	37.2	36.4	32.4	26.7	19.7	13.8		
		MCWB	7.9	10.5	13.8	16.3	20.3	22.6	23.1	23.0	21.2	17.6	12.9	8.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.3	17.6	20.2	21.7	24.2	26.5	26.7	26.1	25.0	23.0	18.9	16.5		
		MCDB	20.4	21.2	25.6	27.3	31.1	35.0	35.1	35.0	33.0	29.4	25.0	19.0		
	2%	WB	12.2	15.1	18.8	20.3	23.2	25.3	25.5	25.2	23.9	21.8	17.2	13.6		
		MCDB	18.0	20.6	24.0	25.7	30.6	33.4	34.5	34.4	31.3	26.9	22.2	16.7		
	5%	WB	10.1	13.1	17.2	19.2	22.5	24.5	24.8	24.5	23.2	20.6	15.8	11.2		
		MCDB	16.3	19.4	21.7	24.4	29.5	32.5	33.7	33.4	30.4	25.5	20.8	15.9		
	10%	WB	8.2	11.1	15.3	17.9	21.6	23.8	24.1	23.9	22.5	19.4	14.1	8.8		
		MCDB	14.3	17.6	20.5	23.0	28.1	31.6	32.7	32.5	29.3	24.9	18.4	13.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	14.9	14.7	14.8	14.9	13.5	13.4	14.0	13.8	13.3	14.5	14.4	13.1		
		MCDBR	20.8	20.7	18.9	19.5	17.0	16.7	16.9	16.6	16.5	17.2	18.4	18.2		
		MCWBR	12.1	11.5	9.7	9.0	6.0	5.0	4.4	4.7	5.2	7.4	9.9	11.0		
		MCDBR	19.2	18.3	14.9	14.7	13.7	13.3	13.8	13.6	13.1	13.1	15.4	15.8		
	5% WB	MCWBR	11.8	11.2	8.5	8.5	6.1	4.9	4.6	4.7	4.7	6.9	9.6	10.7		
(40) Clear-Sky Solar Irradiance	taub		0.283	0.298	0.331	0.365	0.387	0.414	0.416	0.407	0.375	0.343	0.311	0.285		
	taud		2.535	2.497	2.423	2.353	2.362	2.323	2.357	2.329	2.394	2.471	2.490	2.529		
	Ebn at Noon		936	954	943	920	898	870	866	869	884	889	890	911		
	Edh at Noon		80	93	109	123	124	128	123	125	111	95	83	76		
(44) All-Sky Solar Radiation	RadAvg		3.04	3.65	4.86	5.92	6.39	7.03	7.01	6.38	5.32	4.09	3.17	2.58		
	RadStd		0.30	0.48	0.31	0.46	0.66	0.58	0.40	0.37	0.36	0.55	0.39	0.26		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page